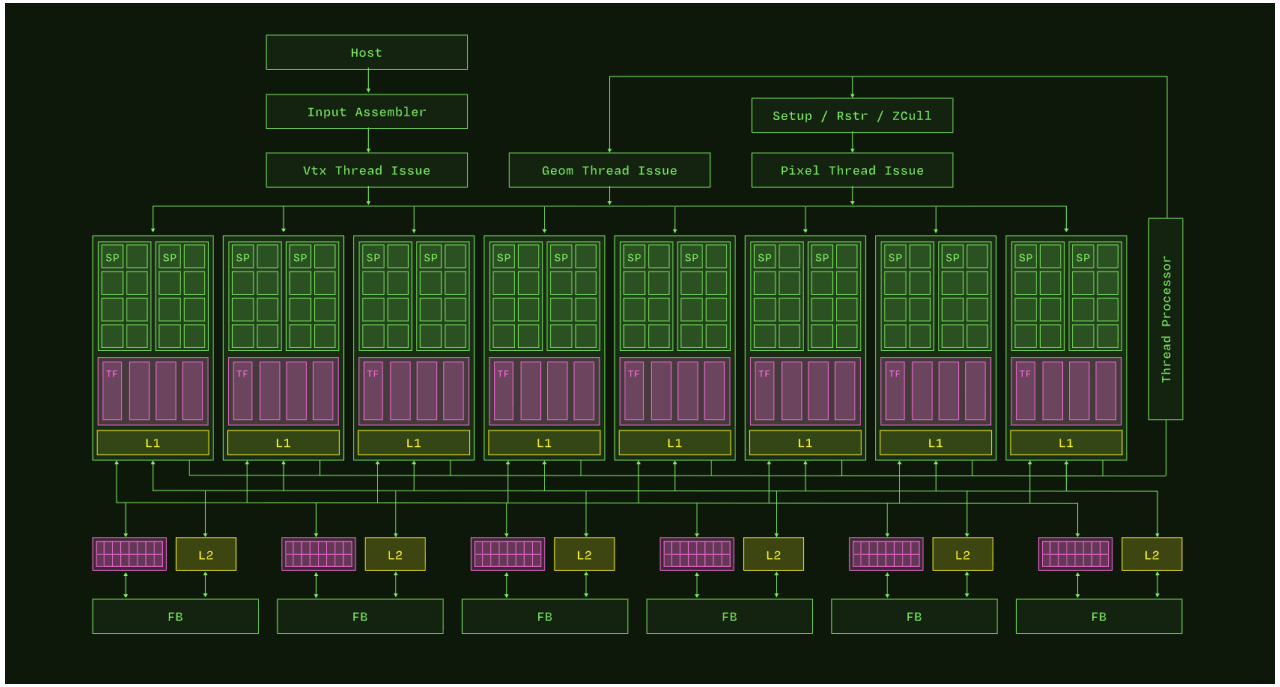
xPU Computer Architecture

AVLAB (Autonomous Vehicle Lab)

양경진 | 2026. 10.07

GPU



Massively Parallel

수천 개 스레드를 동시에 실행

Streaming Multiprocessor

SP 묶음 + L1 메모리 공유

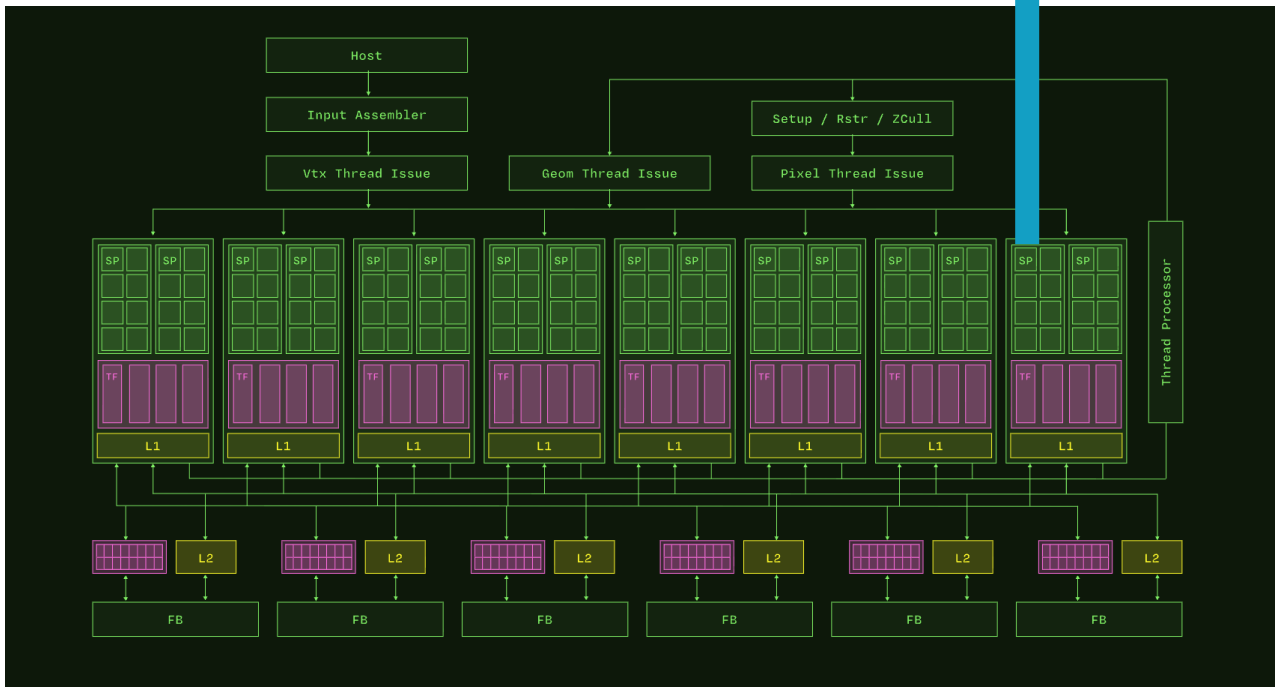
Thread Scheduler

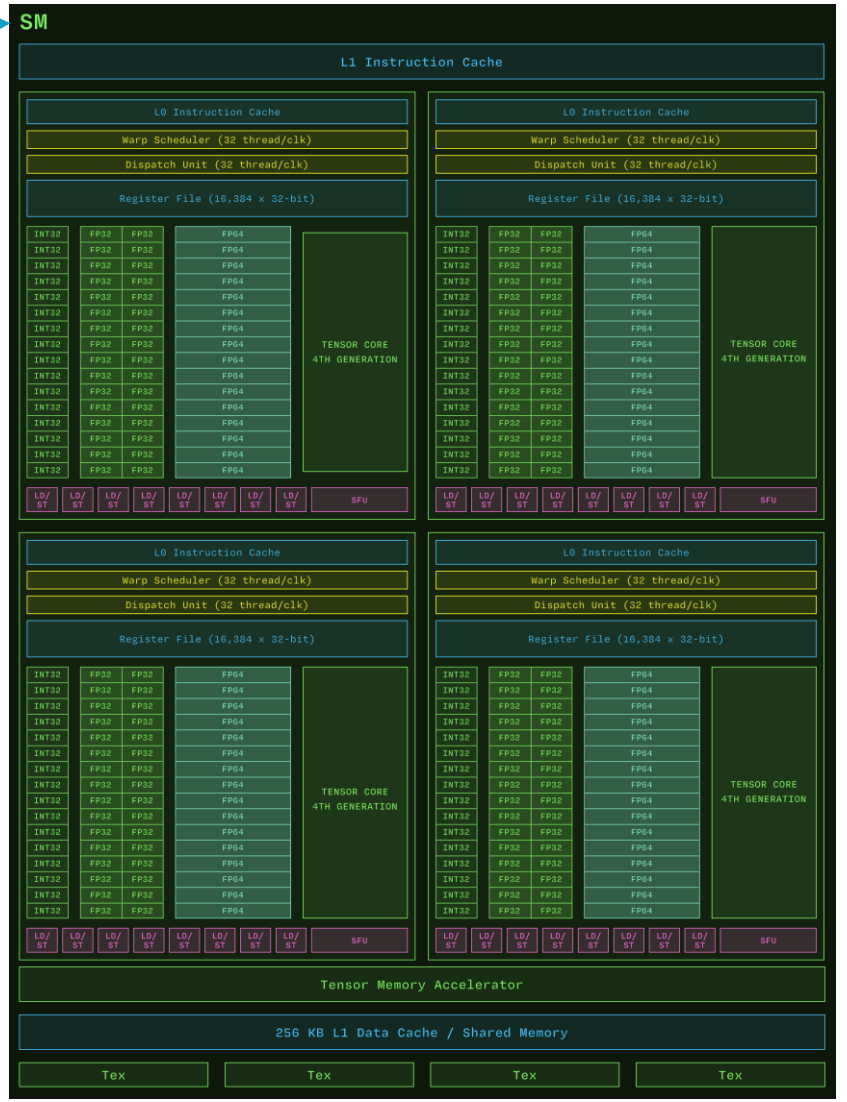
Vertex · Geometry · Pixel 작업 발급

Memory Hierarchy

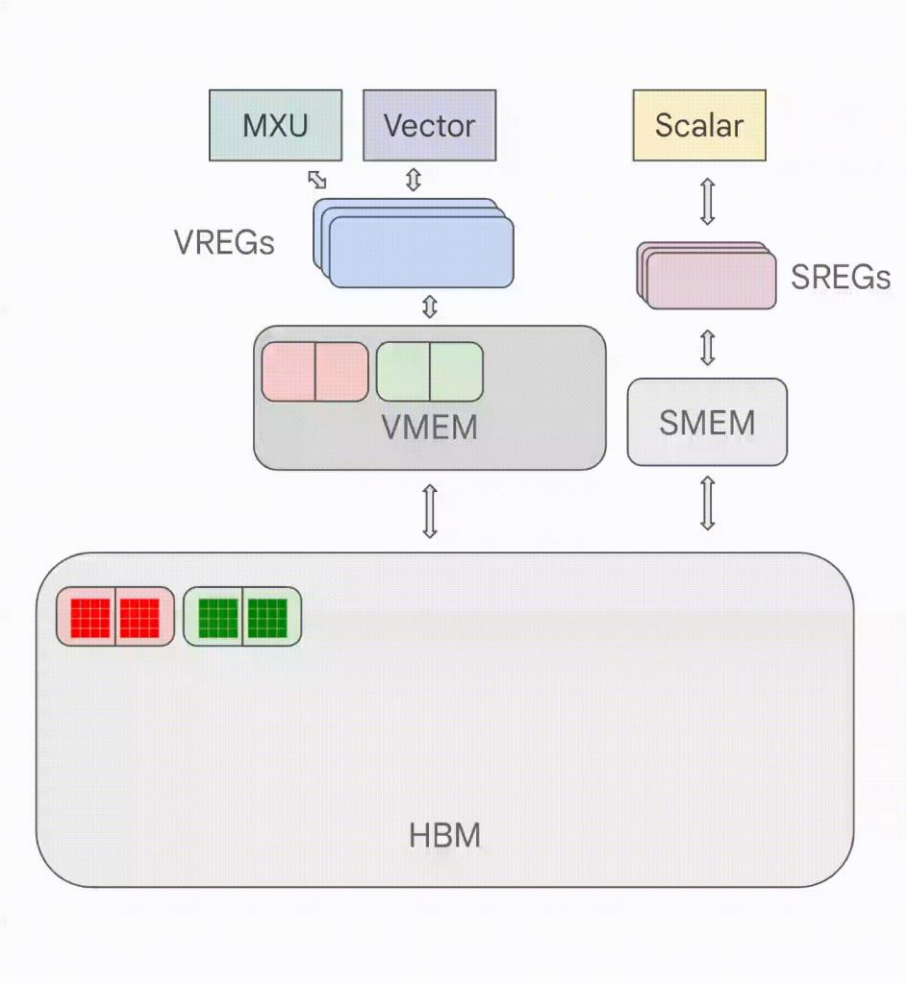
L1 → L2 → Frame Buffer

GPU


Host
Input Assembler
Vtx Thread Issue
Geom Thread Issue
Setup / Rstr / ZCull
Pixel Thread Issue
SP
TF
L1
L2
FB
Thread Processor


SM
L1 Instruction Cache
L0 Instruction Cache
Warp Scheduler (32 thread/clk)
Dispatch Unit (32 thread/clk)
Register File (16,384 x 32-bit)
INT32
FP32
FP64
TENSOR CORE
4TH GENERATION
LD/ST
SFU
Tensor Memory Accelerator
256 KB L1 Data Cache / Shared Memory
Tex

TPU



Systolic Array (MXU)

행렬 곱셈 전용 연산 유닛

Vector · Scalar Unit

요소별 연산과 제어 흐름 처리

On-chip Memory

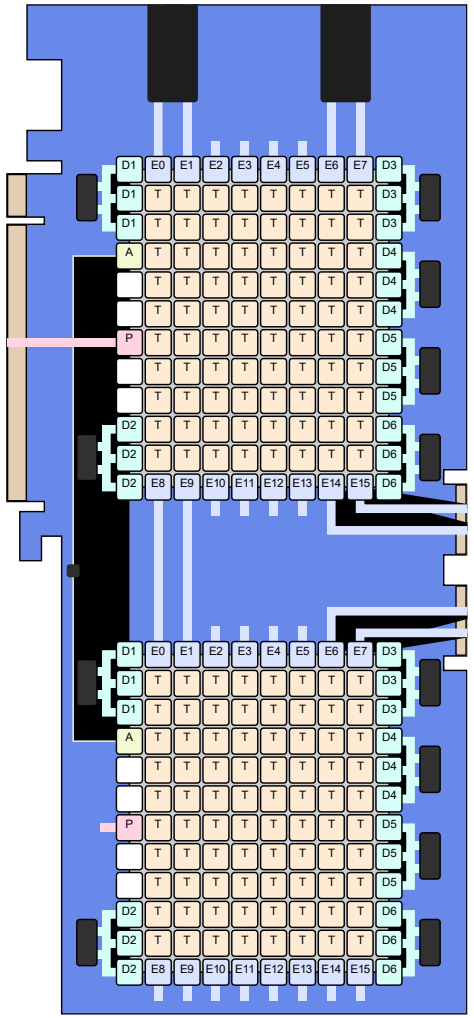
VMEM · SMEM과 VREGs · SREGs

HBM

고대역폭 외부 메모리

Tenstorrent





Wormhole · 칩 2개

한 카드에 Wormhole 칩 2개 탑재

Tensix Core Grid

코어마다 연산 유닛 + 로컬 SRAM

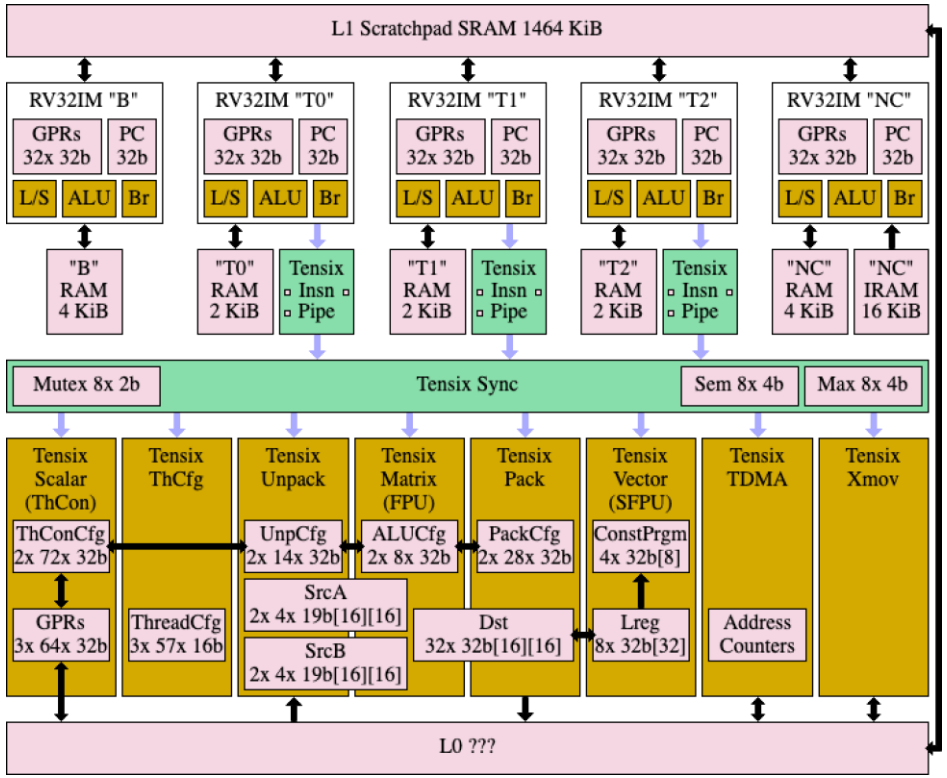
NoC

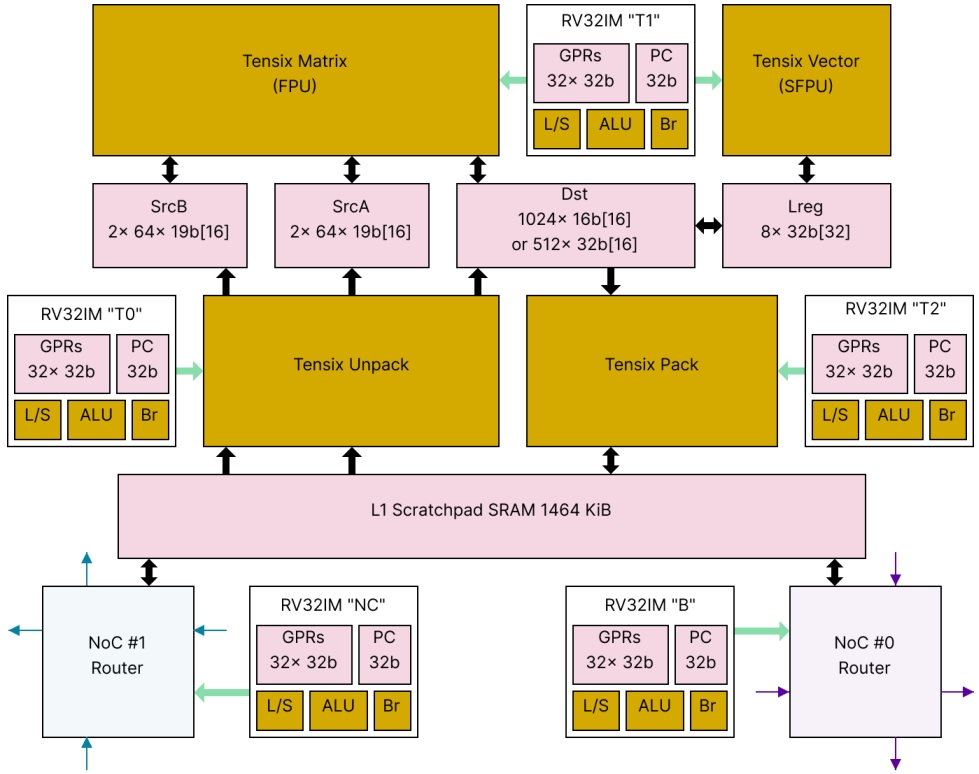
코어 간 데이터를 2D 메시로 전달

Ethernet · GDDR6

칩 간 확장과 외부 메모리

Tenstorrent





THANK YOU
FOR YOUR ATTENTION

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